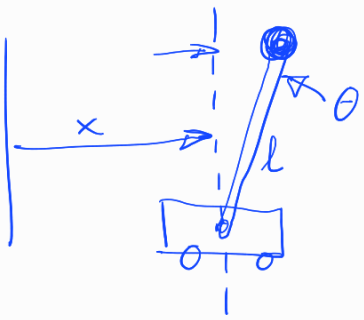
Derive the equations of motion for an inverted pendulum on a rolling cart. Assume $\theta = 0$ is upright and linearize the system for small perturbation around $\theta = 0$



Mass of the pendulum bob is m.
Neglect mass of pendulum arm.
Arm length is l.

open-loop dynamics:

$$X = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} \qquad \dot{X} = AX + Bu$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = A \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} + Bu$$

$\dot{x}_1 = \dot{x} \Rightarrow$ row 1 of A is $[0\ 1\ 0\ 0]$

Treat cart acceleration as the input u.

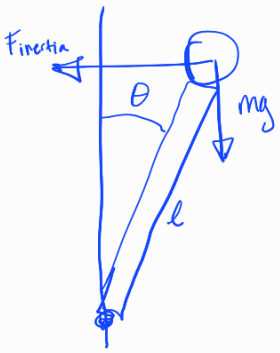
$\Rightarrow \dot{x}_2 = \ddot{x} \Rightarrow B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ ? \end{bmatrix}$ and row 2 of A is $[0\ 0\ 0\ 0]$.

$\dot{x}_3 = \dot{\theta} \Rightarrow$ row 3 of A is $[0\ 0\ 0\ 1]$

So far, $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ ? & ? & ? & ? \end{bmatrix}$

Need to determine $\dot{x}_4 = \ddot{\theta}$

$$\sum \tau_{pivot} = I_{pivot}\ddot{\theta}$$



For a point-mass pendulum, $I_{pivot} = ml^2$

τ_g is the torque due to gravity

τ_c is the torque due to the acceleration of the cart

$\tau_g = mgl\sin\theta$

In the frame of the cart, $F_{inertia}$ on the pedulum $= -m\ddot{x}$

$\Rightarrow \tau_c = F_{inertia}\, l\cos\theta = -m\ddot{x}\, l\cos\theta$

Therefore, $\sum \tau = m l^2 \ddot{\theta} = mgl\sin\theta - m\ddot{x}\, l\cos\theta$

$$l\ddot{\theta} = g\sin\theta - \ddot{x}\cos\theta$$

$$\ddot{\theta} = \frac{g}{l}\sin\theta - \frac{\ddot{x}}{l}\cos\theta$$

Assuming small θ,

$$\ddot{\theta} \approx \frac{g}{l}\theta - \frac{u}{l} \implies \dot{x}_4 = \frac{g x_3}{l} - \frac{u}{l}$$

Therefore,

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{g}{l} & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -\frac{1}{l} \end{bmatrix}$$

Next, derive the LQR closed loop dynamics. $u = -Kx$

$$\dot{x} = (A - BK)x \qquad K = [k_1\ k_2\ k_3\ k_4] \qquad u = -[k_1\ k_2\ k_3\ k_4]\begin{bmatrix} x \\ \dot{x} \\ \theta \\ \ddot{\theta} \end{bmatrix}$$

$$u = -k_1 x - k_2\dot{x} - k_3\theta - k_4\ddot{\theta}$$

We've assumed $u = \ddot{x}$,

$$\ddot{x} = -k_1 x - k_2\dot{x} - k_3\theta - k_4\ddot{\theta}$$

$$J = \int_0^\infty (x^T Q x + u^T R u)\, dt \qquad \text{(infinite horizon)}$$

$$Q = \begin{bmatrix} q_x & 0 & 0 & 0 \\ 0 & q_{\dot{x}} & 0 & 0 \\ 0 & 0 & q_\theta & 0 \\ 0 & 0 & 0 & q_{\ddot{\theta}} \end{bmatrix} \qquad R = r \text{ (scalar)}$$

Algebraic Riccati Equation:

$$A^T P + PA - PBR^{-1}B^T P + Q = 0 \quad , \quad K = R^{-1}B^T P$$